

HESSIAN DISCRIMINANTS AND AUTOMORPHISMS OF BLOW-UPS

OPEN AI

ABSTRACT. Let $\pi: Y = \text{Bl}_S X \rightarrow X$ be the blow-up of a smooth complex projective n -fold along a nonempty connected smooth centre of codimension $c \geq 2$. Write $W = \pi^* N^1(X)_{\mathbf{R}} \subset N^1(Y)_{\mathbf{R}}$ and let Δ_Y be the determinant of the Hessian bilinear form of the top intersection polynomial. We prove that W occurs in the Hessian discriminant with exact multiplicity $c - 2$. For $c \geq 3$, it is therefore a linear component with finite orbit under $\text{Aut}(Y)$. Its stabilizer is exactly the subgroup obtained by lifting $\text{Aut}(X, S)$, and

$$[\text{Aut}(Y) : \text{Aut}(X, S)] \leq \left\lfloor \frac{(n-2)\rho(Y)}{c-2} \right\rfloor.$$

For a finite sequence of blow-ups along smooth centres of codimension at least three, a uniform power of every automorphism descends through the whole sequence. Consequently, if the initial smooth complex projective variety has finite automorphism group, then so does the final blow-up. Thus the examples asked for in Lesieutre–Litt’s Question 1.8 do not occur when the initial variety is smooth over $\mathbf{C}$.

1. INTRODUCTION

Let X be a smooth complex projective variety and let Y be obtained from X by a sequence of ordinary blow-ups along smooth centres. A basic question is whether the birational modification can create new automorphisms. Bayraktar–Cantat obtained strong restrictions when the centres have sufficiently small dimension, and related questions were studied by Truong; see [1, 8]. Lesieutre–Litt proved a uniform descent theorem under the hypotheses

$$2r + 3 \leq n,$$

or $r + 3 \leq n$ together with polyhedrality of the exceptional nef cone.

for a centre of dimension r in an n -fold [6, Theorem 1.6]. They left open the intermediate range and asked whether blow-ups along smooth centres of codimension at least three can turn a finite automorphism group into an infinite one [6, Question 1.8].

The argument below uses only the top intersection polynomial on N^1 . For a smooth projective n -fold Z , set

$$P_Z(h) = \int_Z h^n, \quad q_h(u, v) = \int_Z h^{n-2} uv, \quad \Delta_Z(h) = \det(q_h).$$

2020 *Mathematics Subject Classification.* 14J50, 14E05, 14C17.

Key words and phrases. blow-up, automorphism group, intersection form, Hessian discriminant, exceptional divisor.

Up to a nonzero scalar, Δ_Z is the determinant of the ordinary Hessian of P_Z . If $\pi: Y = \text{Bl}_S X \rightarrow X$ has centre of codimension c , then the pullback space

$$W = \pi^* N^1(X)_{\mathbf{R}}$$

is a hyperplane in $N^1(Y)_{\mathbf{R}}$. The key point is the exact factorization order

$$\text{ord}_W(\Delta_Y) = c - 2. \quad (1)$$

Thus codimension three is exactly the first codimension for which the blow-down is recorded by a linear component of the Hessian discriminant. Automorphisms preserve Δ_Y up to a nonzero scalar, hence permute these linear components with their multiplicities. The orbit of W is finite, and its stabilizer turns out to be exactly the subgroup lifted from automorphisms of the pair (X, S) .

Intersection forms have long been used to constrain birational geometry. In dimension three, Bisi–Cascini–Tasin used the cubic intersection form to obstruct smooth blow-up presentations [2]. More recently, they studied the hypermatrix of the $(n-1)$ st derivatives of the full intersection form; in particular, they derived a blow-up expansion and used low-rank exceptional classes to constrain divisorial contractions [3, Sections 4–5]. The invariant used here is different: it is the determinant of the ordinary Hessian on N^1 , and the relevant locus is the pullback hyperplane W , rather than the exceptional class itself.

The exact multiplicity in (1) yields both an explicit index bound and a uniform iterate theorem. The latter removes the dimension and nef-cone hypotheses from Lesieutre–Litt’s descent theorem when every centre has codimension at least three.

2. THE HESSIAN DISCRIMINANT

Let Z be a smooth complex projective n -fold, $n \geq 3$, and put

$$V_Z = N^1(Z)_{\mathbf{R}}, \quad \rho(Z) = \dim_{\mathbf{R}} V_Z.$$

Choose a basis $D_1, \dots, D_\rho$ of V_Z . Define

$$q_h(u, v) = \int_Z h^{n-2} uv, \quad M_Z(h) = (q_h(D_i, D_j))_{i,j}, \quad \Delta_Z(h) = \det M_Z(h). \quad (2)$$

If $h = \sum x_i D_i$, then

$$\det \text{Hess}(P_Z) = (n(n-1))^\rho \Delta_Z.$$

A change of basis multiplies Δ_Z by a nonzero square. Hence the zero hypersurface of Δ_Z , and the multiplicity of each real linear factor, are intrinsic.

Lemma 2.1. *The polynomial Δ_Z is nonzero and homogeneous of degree $(n-2)\rho(Z)$. Its zero hypersurface is preserved by $\text{Aut}(Z)$, including the multiplicity of every irreducible factor.*

Proof. For an ample class a , the Hodge index theorem gives signature $(1, \rho(Z) - 1)$ for

$$(u, v) \mapsto \int_Z a^{n-2} uv$$

on $N^1(Z)_{\mathbf{R}}$; see, for example, [4, Theorem 7.3.4]. Hence $\Delta_Z(a) \neq 0$. Each entry of $M_Z(h)$ is homogeneous of degree $n - 2$, so $\deg \Delta_Z = (n - 2)\rho(Z)$.

If $g \in \text{Aut}(Z)$ and T is the matrix of g^* in the chosen basis, invariance of intersection numbers gives

$$T^t M_Z(g^* h) T = M_Z(h).$$

Thus

$$\Delta_Z(g^* h) = (\det T)^{-2} \Delta_Z(h).$$

Pullback therefore preserves the zero hypersurface of Δ_Z and all factor multiplicities. $\square$

3. THE MULTIPLICITY OF THE BLOW-DOWN HYPERPLANE

Let

$$\pi: Y = \text{Bl}_S X \longrightarrow X$$

be the blow-up of a smooth complex projective n -fold along a nonempty connected smooth centre S of codimension $c \geq 2$. Let E be the exceptional divisor. Then

$$N^1(Y)_{\mathbf{R}} = \pi^* N^1(X)_{\mathbf{R}} \oplus \mathbf{R}[E]. \quad (3)$$

Write

$$W = \pi^* N^1(X)_{\mathbf{R}}.$$

We first record the only intersection calculation needed below.

Lemma 3.1. *For $1 \leq j < c$ one has $\pi_* E^j = 0$, while*

$$\pi_* E^c = (-1)^{c-1} [S].$$

Proof. Let $p: E = \mathbf{P}(N_{S/X}) \rightarrow S$ and put $\xi = c_1(\mathcal{O}_E(1))$. Since $\mathcal{O}_E(E) = \mathcal{O}_E(-1)$,

$$E^j = i_* ((-\xi)^{j-1}).$$

The projective bundle formula gives $p_* \xi^k = 0$ for $k < c - 1$ and $p_* \xi^{c-1} = 1$. Pushing forward through $S \hookrightarrow X$ proves the claim; compare [5, Chapter 3]. $\square$

Theorem 3.2 (Exact Hessian multiplicity). *With the notation above,*

$$\boxed{\text{ord}_W(\Delta_Y) = c - 2.}$$

Equivalently, if ℓ_W is a linear equation for W , then

$$\ell_W^{c-2} \mid \Delta_Y, \quad \ell_W^{c-1} \nmid \Delta_Y.$$

In particular, W is a component of the Hessian discriminant if and only if $c \geq 3$.

Proof. Choose a basis $D_1, \dots, D_r$ of $N^1(X)_{\mathbf{R}}$ and use

$$\pi^* D_1, \dots, \pi^* D_r, E$$

as a basis of $N^1(Y)_{\mathbf{R}}$. Write

$$h = \pi^* a + tE.$$

Relative to this basis, the matrix of q_h has block form

$$M_Y(h) = \begin{pmatrix} A(t) & b(t) \\ b(t)^\dagger & d(t) \end{pmatrix}.$$

By lemma 3.1 and the projection formula,

$$A(t) = A(0) + O(t^c), \quad (4)$$

$$b(t) = O(t^{c-1}), \quad (5)$$

$$d(t) = (-1)^{c-1} \binom{n-2}{c-2} \left(\int_S (a|_S)^{n-c} \right) t^{c-2} + O(t^{c-1}). \quad (6)$$

The block estimates hold in the polynomial ring in the coordinates of a and t . Expanding the determinant by permutations therefore shows directly that

$$t^{c-2} \mid \det M_Y(\pi^*a + tE)$$

as a polynomial, not merely after specializing a .

Now take a ample. The Hodge index theorem gives $\det A(0) \neq 0$, and ampleness on S gives

$$\int_S (a|_S)^{n-c} > 0.$$

In the formal power series ring at $t = 0$,

$$\det M_Y(h) = \det A(t) (d(t) - b(t)^\dagger A(t)^{-1} b(t)).$$

The second term in the parentheses has order at least $2c - 2$, whereas (6) has a nonzero term of order $c - 2$. Therefore the coefficient of t^{c-2} in $\det M_Y(h)$ is nonzero for this ample a . The global divisibility by t^{c-2} is consequently exact. $\square$

Remark 3.3 (The codimension-two boundary). For $c = 2$, theorem 3.2 gives multiplicity zero. More concretely, for ample a the block $A(0)$ is nondegenerate, the mixed block vanishes at $t = 0$, and

$$d(0) = - \int_S (a|_S)^{n-2} < 0.$$

Hence $\Delta_Y(\pi^*a) \neq 0$. This is exactly where the Hessian-hyperplane mechanism stops. Lesieutre–Litt exhibit codimension-two blow-ups for which uniform descent fails [6, Example 2.5].

4. AUTOMORPHISMS OF A SINGLE BLOW-UP

Write

$$\text{Aut}(X, S) = \{\varphi \in \text{Aut}(X) : \varphi(S) = S\}.$$

Every element of $\text{Aut}(X, S)$ lifts uniquely to Y .

Theorem 4.1 (Virtual recovery of the blow-down). *Assume $c \geq 3$ and set*

$$M(Y, S) = \left\lfloor \frac{(n-2)\rho(Y)}{c-2} \right\rfloor.$$

Then the natural lift identifies $\text{Aut}(X, S)$ with the stabilizer of W in $\text{Aut}(Y)$. In particular,

$$[\text{Aut}(Y) : \text{Aut}(X, S)] \leq M(Y, S). \quad (7)$$

Moreover, if

$$L(Y, S) = \text{lcm}(1, 2, \dots, M(Y, S)),$$

then for every $g \in \text{Aut}(Y)$ the iterate $g^{L(Y, S)}$ descends to an automorphism of X preserving S .

Proof. By theorem 3.2, the linear form defining W divides Δ_Y with multiplicity $c - 2$. By lemma 2.1, $\text{Aut}(Y)$ permutes the linear factors having this multiplicity. There are at most

$$\left\lfloor \frac{\deg \Delta_Y}{c - 2} \right\rfloor = M(Y, S).$$

Thus the stabilizer

$$G_W = \{g \in \text{Aut}(Y) : g^*W = W\}$$

has index at most $M(Y, S)$.

It remains to identify G_W . Let $g \in G_W$ and choose an ample divisor A on X . If $C \subset Y$ is an integral curve contracted by π , then

$$A \cdot \pi_* g_* [C] = (g^* \pi^* A) \cdot C = 0,$$

because $g^* \pi^* A \in W$. If $\pi(g(C))$ were a curve, ampleness would make the left side positive. Hence g sends every contracted curve to a contracted curve.

Each nontrivial fibre of π is $\mathbf{P}^{c-1}$, whose closed points are joined by lines. Therefore πg is constant on every fibre. Let Γ be the reduced image of

$$(\pi, \pi g) : Y \longrightarrow X \times X.$$

This map is proper, so its image is closed. The first projection $q : \Gamma \rightarrow X$ is proper. For every $x \in X$, all points of $\pi^{-1}(x)$ have the same image in Γ ; since $Y \rightarrow \Gamma$ is surjective, the underlying topological space of $q^{-1}(x)$ consists of one point. Thus q has finite fibres and is finite [7, Tag 02LS]. Over $X \setminus S$ it is the graph of the morphism $\pi g \pi^{-1}$, so q is birational. The variety Γ is integral, and X is normal; hence q is an isomorphism [7, Tag 0AB1]. The second projection therefore gives a morphism $\bar{g} : X \rightarrow X$ satisfying

$$\pi g = \bar{g} \pi.$$

Applying the same argument to g^{-1} gives a morphism $\overline{g^{-1}} : X \rightarrow X$. Since π is surjective,

$$\bar{g} \overline{g^{-1}} = \overline{g^{-1}} \bar{g} = \text{id}_X,$$

so $\bar{g} \in \text{Aut}(X)$.

For $y \in E$, choose a line in the fibre of $E \rightarrow S$ through y . Its image under g is contracted by π . Every curve contracted by π is contained in E , because π is an isomorphism over $X \setminus S$. Hence $g(E) \subseteq E$; since both are irreducible divisors, $g(E) = E$. Therefore

$$\bar{g}(S) = \pi g(E) = S.$$

If $\bar{g} = \text{id}_X$, then g is the identity on $Y \setminus E$, and hence on Y . Therefore descent gives an injection $G_W \hookrightarrow \text{Aut}(X, S)$. Conversely, every automorphism of (X, S) lifts uniquely to the blow-up by its universal property, and the lift preserves W . Hence

$$G_W \cong \text{Aut}(X, S),$$

which proves (7).

Finally, the orbit of W under a fixed g has length at most $M(Y, S)$. Therefore g^m stabilizes W for some $1 \leq m \leq M(Y, S)$, and $g^{L(Y, S)}$ does so uniformly for every g . $\square$

5. SEQUENCES OF BLOW-UPS

A blow-up along a finite disjoint union of smooth connected centres is the same as the successive blow-up of the components. We therefore state the result for connected centres.

Theorem 5.1 (Uniform descent). *Let*

$$Y = X_m \xrightarrow{\pi_m} X_{m-1} \longrightarrow \cdots \xrightarrow{\pi_1} X_0 = X$$

be a finite sequence of smooth complex projective n -folds, where π_i is the blow-up along a nonempty connected smooth centre $S_i \subset X_{i-1}$ of codimension $c_i \geq 3$. Put

$$M_i = \left\lfloor \frac{(n-2)\rho(X_i)}{c_i-2} \right\rfloor, \quad L_i = \text{lcm}(1, \dots, M_i), \quad N = \prod_{i=1}^m L_i.$$

Then:

- (i) *there is a finite-index subgroup $G \leq \text{Aut}(Y)$ that descends faithfully to $\text{Aut}(X)$;*
- (ii)

$$[\text{Aut}(Y) : G] \leq \prod_{i=1}^m M_i;$$

- (iii) *for every $g \in \text{Aut}(Y)$, the iterate g^N descends to an automorphism of X .*

Consequently, if $\text{Aut}(X)$ is finite, then $\text{Aut}(Y)$ is finite.

Proof. For each i , theorem 4.1 gives a subgroup $G_i \leq \text{Aut}(X_i)$ of index at most M_i together with an injective descent homomorphism

$$\delta_i : G_i \longrightarrow \text{Aut}(X_{i-1}).$$

Define recursively $H_0 = \text{Aut}(X_0)$ and

$$H_i = \delta_i^{-1}(H_{i-1}) \subset G_i.$$

Then

$$[G_i : H_i] = [\delta_i(G_i) : \delta_i(G_i) \cap H_{i-1}] \leq [\text{Aut}(X_{i-1}) : H_{i-1}].$$

Induction gives

$$[\text{Aut}(X_i) : H_i] \leq \prod_{j=1}^i M_j.$$

Taking $G = H_m$ proves (i) and (ii), and faithfulness follows from the single-step faithfulness at every stage.

For (iii), start with $g_m = g$. By theorem 4.1, $g_m^{L_m}$ descends to some $g_{m-1} \in \text{Aut}(X_{m-1})$. Then $g_{m-1}^{L_{m-1}}$ descends to X_{m-2} , and so on. After m steps, g^N descends to X . If $\text{Aut}(X)$ is finite, then G is finite by injectivity; since G has finite index, so is $\text{Aut}(Y)$. $\square$

Corollary 5.2. *Let X be smooth and complex projective. For any finite sequence of blow-ups of X along smooth centres of codimension at least three, Lesieutre–Litt’s uniform-iterate conclusion holds with no additional dimension inequality and no polyhedral-nef-cone hypothesis. In particular, no such sequence produces an infinite automorphism group from a finite one; this rules out the examples sought in [6, Question 1.8] when the initial variety is smooth over $\mathbf{C}$.*

Remark 5.3. The proof is numerical except for the nondegeneracy of the Hodge-index form used in lemma 2.1. No dynamical Mordell–Lang input is required. We have kept the statement over $\mathbf{C}$ in order to isolate the argument from additional characteristic-dependent hypotheses.

AI ASSISTANCE DISCLOSURE

GPT-6 Astra and GPT-5.6 Sol were used during parts of the investigation, drafting, and checking of this note.

REFERENCES

- [1] T. Bayraktar and S. Cantat, *Constraints on automorphism groups of higher dimensional manifolds*, J. Math. Anal. Appl. **405** (2013), no. 1, 209–213.
- [2] C. Bisi, P. Cascini, and L. Tasin, *A remark on the Ueno–Campana’s threefold*, Michigan Math. J. **65** (2016), no. 3, 567–572.
- [3] C. Bisi, P. Cascini, and L. Tasin, *Forms and complex manifolds*, EMS Surv. Math. Sci. (2025), published online first, DOI: 10.4171/EMSS/109.
- [4] M. A. A. de Cataldo, *Lectures on the Hodge theory of projective manifolds*, arXiv:math/0504561 (2005).
- [5] W. Fulton, *Intersection Theory*, 2nd ed., Springer, 1998.
- [6] J. Lesieutre and D. Litt, *Dynamical Mordell–Lang and automorphisms of blow-ups*, Algebraic Geometry **6** (2019), no. 1, 1–25.
- [7] The Stacks Project Authors, *The Stacks Project*, <https://stacks.math.columbia.edu>, Tags 02LS and 0AB1.
- [8] T. T. Truong, *On automorphisms of blowups of projective manifolds*, arXiv:1301.4957 (2013).